

ELEC4631 Tutorial 7

1. Consider again the second order state-space system from Tutorial 6 Q4.
 - (a) Using the output $y(t)$ design an observer for the system so that $A - LC$ has closed-loop eigenvalues at $-4 \pm 3j$.
 - (b) Put together the observer from part (a) and the LQR controller from Tutorial 7 Q4(b) to form an output feedback controller. Sketch a block diagram of the closed-loop system indicating the system, the observer, and the feedback control signal.
2. Consider the system described by the following ODE:

$$\begin{aligned}\dot{x}_1 &= -x_1 - x_2 \\ \dot{x}_2 &= -x_1 - x_2^3.\end{aligned}$$

Find all equilibrium points of this system and using the indirect method of Lyapunov determine the stability property of each equilibrium point.

3. Revisit the damped pendulum from Lecture 5. Using the indirect method of Lyapunov, show that the equilibrium point $(\pi, 0)$ (pendulum is stationary in the upward position) is unstable while the equilibrium point $(0, 0)$ (pendulum is stationary in the downward position) is asymptotically stable.
4. Consider a special case of a nonlinear dynamical system known as the *Rössler system*. The system is described by the nonlinear ODE

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -x_2 - x_3 \\ x_1 - cx_2/2 \\ x_3(x_1 - c) - c/2 \end{bmatrix},$$

for some real constant parameter $c \neq 0$. Here the vector field is:

$$f(x_1, x_2, x_3) = \begin{bmatrix} -x_2 - x_3 \\ x_1 - cx_2/2 \\ x_3(x_1 - c) - c/2 \end{bmatrix}.$$

- (a) Find *all* equilibrium points of this system.
 - (b) Compute the Jacobian $\text{Jac}_x(f)$.
 - (c) Show that if $x^* = (x_1^*, x_2^*, x_3^*)$ is an equilibrium point and $A = \text{Jac}_{x^*}(f)$ then
$$c_A(s) = (s - x_1^* + c)(s(s + c/2) + 1 + x_3^*).$$
 - (d) Determine the stability property of every equilibrium point x^* , if possible, when $c = 1$ using the first/indirect method of Lyapunov.
5. An electronic unity feedback loop is formed with a controller C and plant P in the feedforward loop, see Figure 1. The plant P is an inverting integrator (e.g., realised

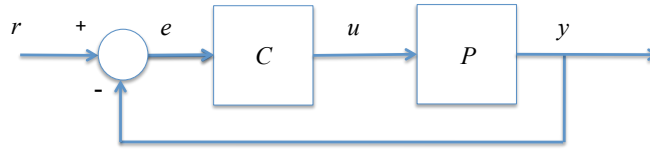


Figure 1: A unity feedback loop.

with an op-amp circuit) with a feedthrough of the input. It has the control signal $u(t)$ as input and $y(t)$ as output. The state-space equation for the plant P is

$$\begin{aligned}\dot{x}_1(t) &= u(t), \\ y(t) &= -x_1(t) + u(t),\end{aligned}$$

and the transfer function of the plant is $P(s) = \frac{s-1}{s}$. The controller has input $e(t)$ (the error signal) and outputs the control signal $u(t)$. It can be implemented with one integrator and has the state-space equation

$$\begin{aligned}\dot{x}_2(t) &= x_2(t) + e(t), \\ u(t) &= x_2(t),\end{aligned}$$

and transfer function $C(s) = \frac{1}{s-1}$. The controller has one pole in the RHP.

The unity feedback loop is driven by a constant set-point r . Questions:

- (a) Show that the closed-loop transfer function $H_{\text{cl}}(s)$ is $H_{\text{cl}}(s) = \frac{1}{s+1}$. Therefore, $y(t)$ can track r as $t \rightarrow \infty$, i.e., $y(t) \rightarrow r$ as $t \rightarrow \infty$ (from ELEC3114).
- (b) Write down the state-space equations for the open-loop feedforward system formed by cascading C and P (the output of C becomes input of P) with $y(t)$ as output. Show that the cascade is controllable but not observable.

- (c) Write down the state-space equation for the closed-loop system with output $y(t)$. Show that this closed-loop system is controllable but not observable.
- (d) Solve the closed-loop equation to show that $y(t) \rightarrow r$ as $t \rightarrow \infty$ yet all is *not* well with the system. That is, for certain initial conditions, $|x_1(t)| \rightarrow \infty$ and $|x_2(t)| \rightarrow \infty$ as $t \rightarrow \infty$.